# Modular Image Processing

Built with Python, PySide6, and the Scientific Stack. A robust, plugin-oriented application for advanced visual transformations.

**NIZAR MODULE INTEGRATION**

# Digital Canvas Theory

Digital images are represented as numerical matrices where each element (pixel) denotes intensity or color value.

**Spatial Domain:** Directly manipulating pixel values based on their position.

**Frequency Domain:** Transforming the image to analyze periodic patterns, often using Fourier Transforms.

Filters act as mathematical operators that redistribute these values to enhance, detect, or distort visual information.

# Modular Architecture

### Plugin Discovery

The app scans the `/modules` folder at runtime, enabling "hot-swappable" processing logic without core modifications.

### Inheritance

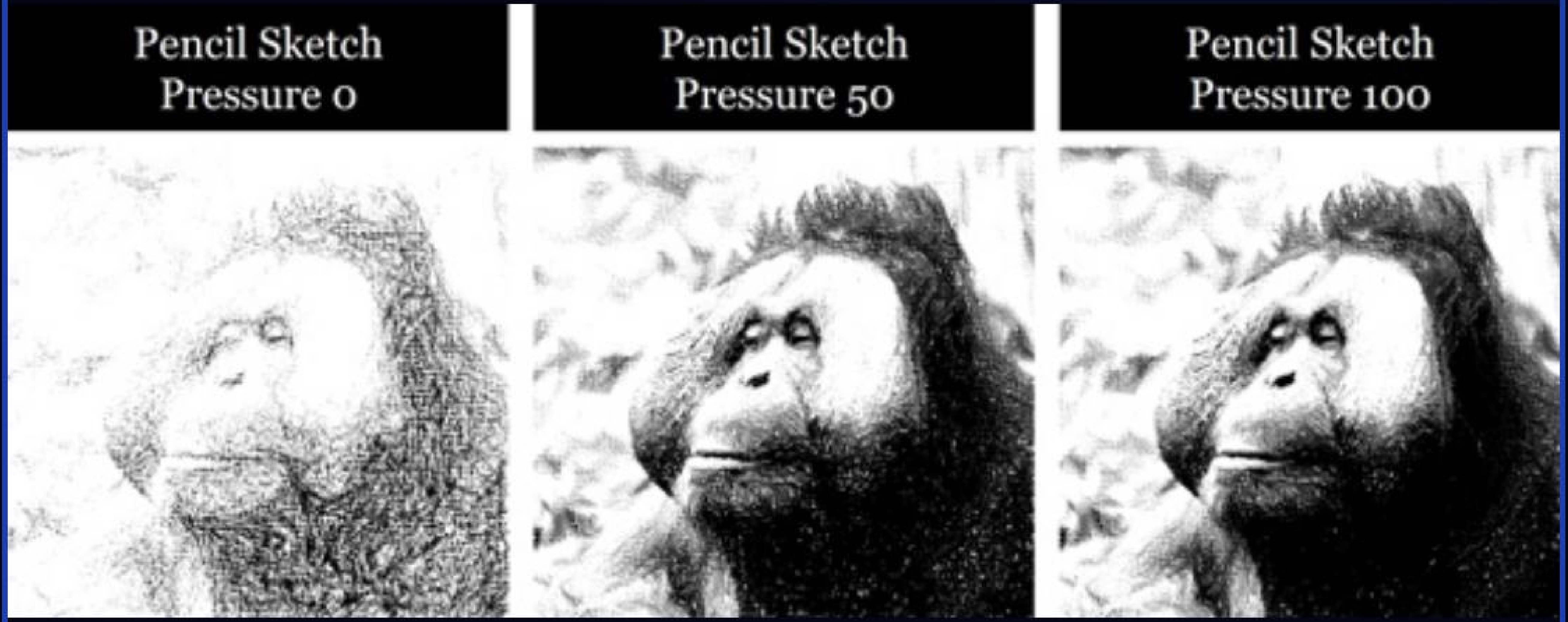
Every module inherits from `IImageModule`, enforcing a strict contract for inputs, outputs, and parameter registration.

### Auto-Registration

Dynamic imports allow the system to automatically register and generate UI controls for any valid module found in the system path.

# The Nizar Suite

- **Sobel Edge:** Gradient magnitude detection.
- **Gaussian Blur:** Weighted smoothing via Sigma control.
- **Power Law (Gamma):** Non-linear contrast correction.
- **Stretching:** Linear intensity rescaling to new ranges.
- **Custom Convolution:** User-defined 3x3 kernel matrices.



# Sketch & Emboss Logic

## Pencil Sketch Effect

A multi-step pipeline: Grayscale → Inversion → Gaussian Blur → Inversion → Dodge-Blend Divide.

$$R = \frac{I_{gray}}{255 - I_{blur}} \times 255$$

This ratio creates high-contrast edges mimicking graphite strokes.

## 3D Emboss Filter

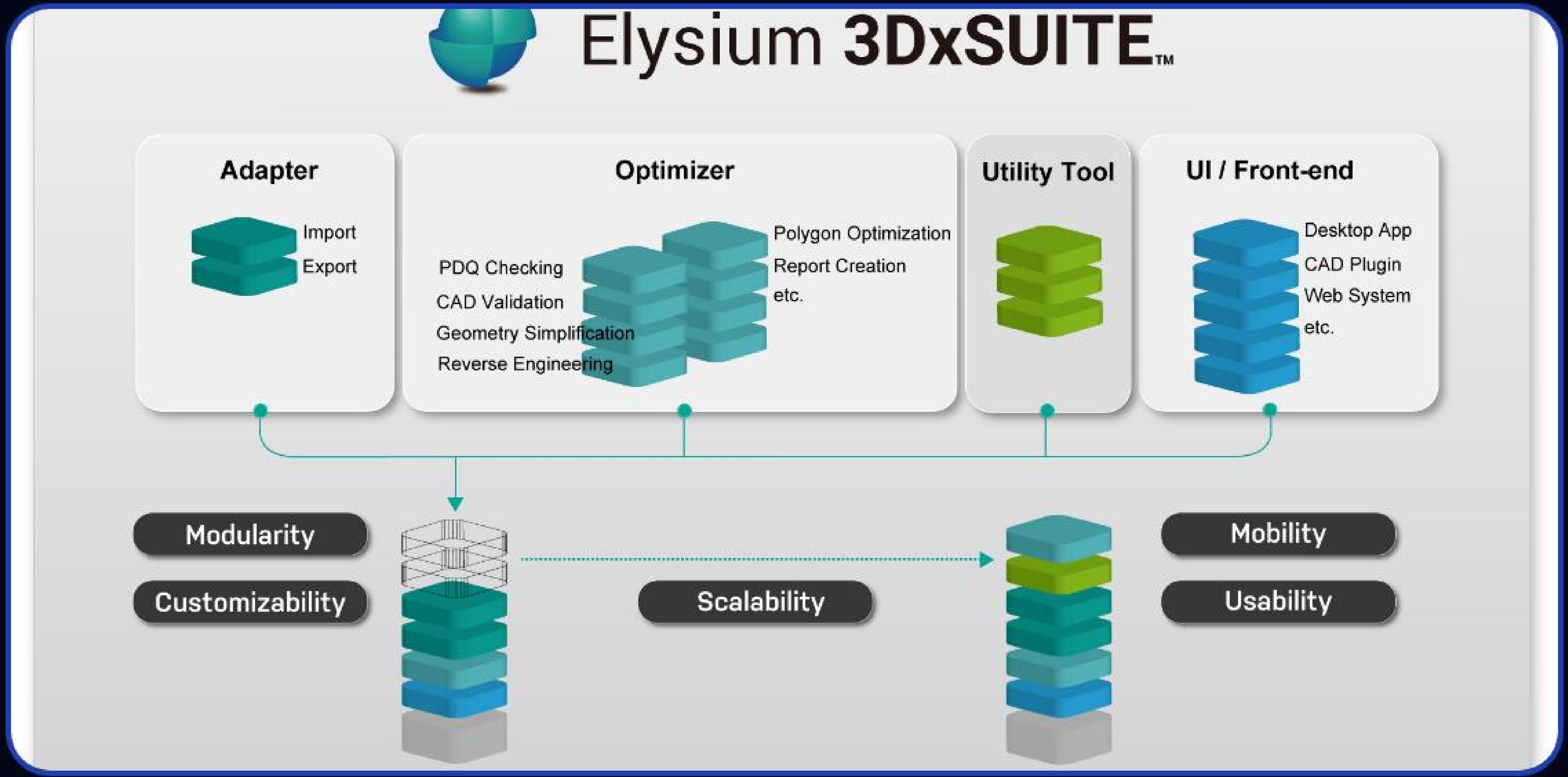
Directional kernels accentuate gradients based on light source direction (Top-Left, Bottom-Right, etc.).

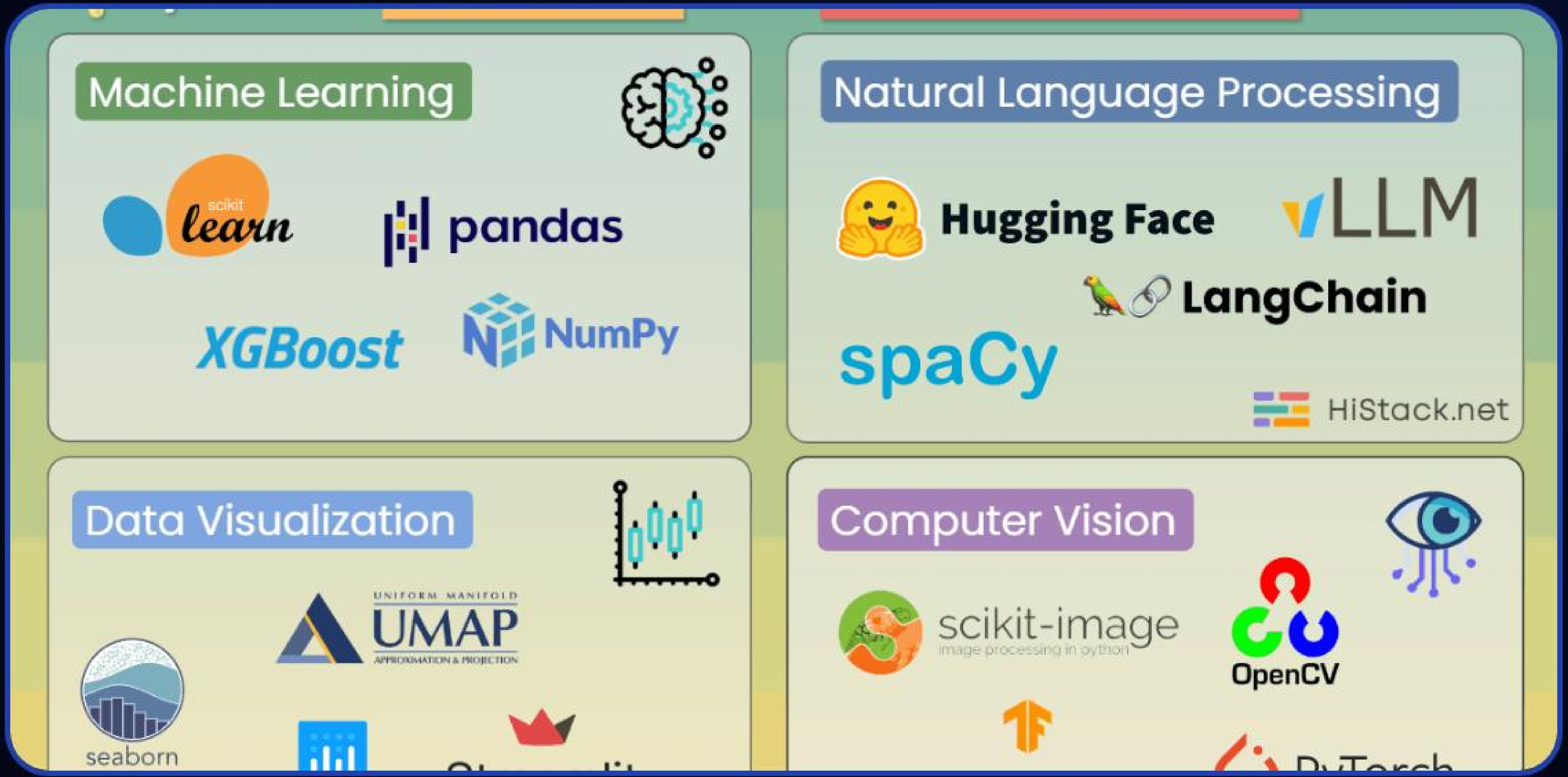
Example 3x3 Kernel (Top-Left):

$$\begin{matrix} -2 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{matrix}$$

Output values are offset by a mid-gray bias to reveal 3D relief.

# Stack & Interface




## Control Panel

Dynamic parameter widgets built with PySide6 for real-time slider updates.

## Core Engine

High-performance processing powered by **NumPy** matrices and **SciPy** signals.

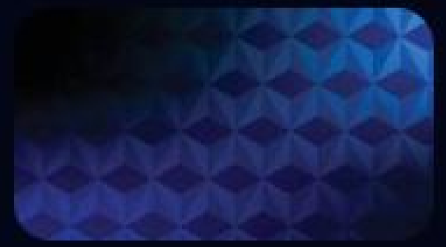
# Thank You!

Empowering digital imagery through Pythonic modularity.

>_ python app_main.py

developer@nizar-module.local

# Image Sources



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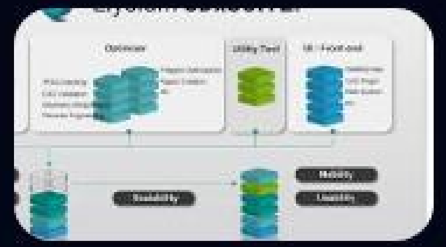
Source: www.vecteezy.com

https://nutsandboltsspeedtraining.com/wp-content/uploads/2018/06/Photo-to-Sketch-10.jpg

Source: nutsandboltsspeedtraining.com



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Source: www.histack.net